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UNIVERSITI TUNKU ABDUL RAHMAN

FACULTY OF INFORMATION & COMMUNICATION TECHNOLOGY

BACHELOR OF INFORMATION TECHNOLOGY (HONS)

COMPUTER ENGINEERING

UCCE1064 PROCESSOR AND INTERFACING

TITLE: Servo Motor Control Module

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1.0 Introduction

1.1 General Information

A servo motor is a motor equipped with a control circuit that allows for precise angular positioning (Apoorve, 2025). This Servo Motor Control Module project is designed to control a servo motor using an 8051 microcontroller development board. The design allows users to select different speed modes via a keypad. Upon selection, the user is prompted to confirm or reset. During operation, an LCD displays the current speed mode, while a 7-segment display provides real-time feedback on the number of completed cycles. The motor moves back and forth with predefined delays based on the selected speed. It completes 10 cycles before stopping. The system also features push buttons for an emergency stop and a reset function.

The system also features push buttons for an emergency stop—which instantly halts all motor and counting operations—and a reset function that returns the LCD to the main menu without interrupting the current motor cycle.

1.2 Product Specifications

Functionalities provided by this product

- LCD Menu Interface: Displays speed options (0, 1, 2, 3), prompts for confirmation and shows real-time running status
- Speed Mode Selection: Wait times for backward and forward movements:
 - Speed 0: Wait 3 s for single direction (forward or backward)
 - Speed 1: Wait 2.5 s for single direction (forward or backward)
 - Speed 2: Wait 2 s for single direction (forward or backward)
 - Speed 3: Wait 1 s for single direction (forward or backward)
- Cycle Counter: Counts up to 10 completed back-and-forth cycles using a 7-segment display
- Emergency Stop: Immediately stops the servo motor and returns to the menu when the designated push button is pressed
- Reset: Returns the system to the menu without stopping the running servo motor when the designated push button is pressed

2.0 Hardware and Software Partition

2.1 Hardware Partition

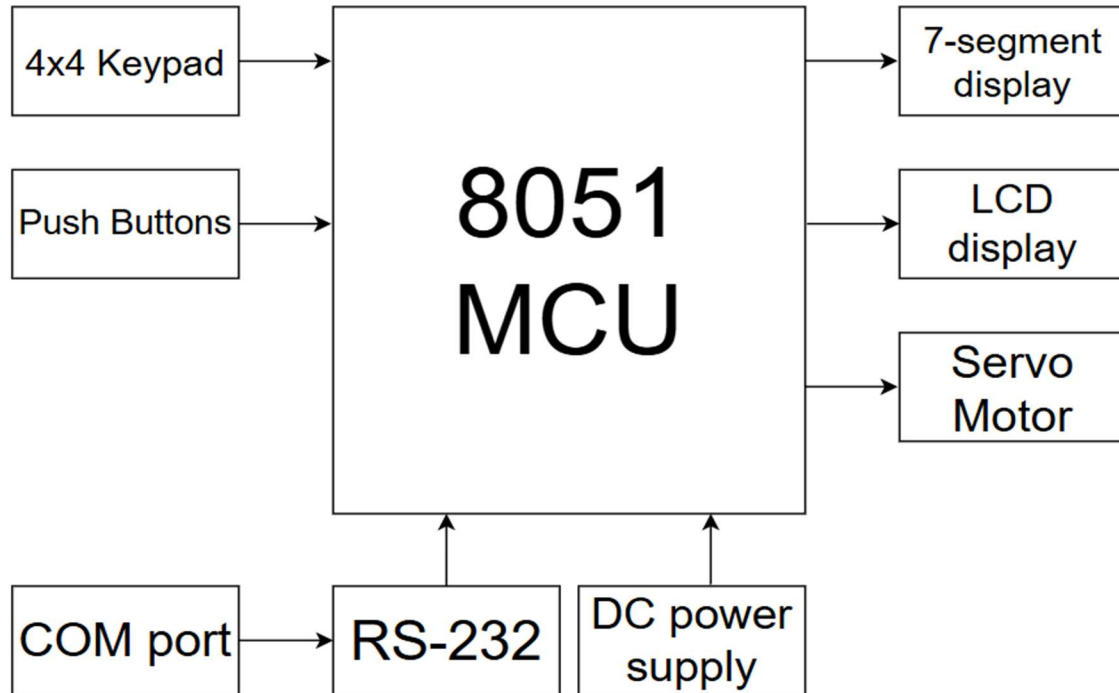


Figure 2.1.1 Block diagram of Servo Motor Control Module

No.	Block	Description
1	8051 MCU	Interfaces with the keypad, LCD, servo motor, 7-segment display, and push buttons. Connections: P0 → LCD (LCD_DAT) P1 → 4x4 Keypad (KEY_PORT) P2 → 7-segment display data (SEG_PORT) P3.0 → LCD Enable (LCD_EN) P3.1 → LCD Read/Write (LCD_RW) P3.4 → LCD Register Select (LCD_RS) P3.5 → Servo Motor PWM (SERVO_PIN) P3.6, P3.7 → 7-segment multiplexing (DIS0, DIS1)

2	4x4 Keypad	Allows the user to select speed modes (0-3) and confirm/cancel (A/B). Connection: 8051 MCU (P1)
3	Push Buttons	Used for hardware interrupts (Emergency Stop and Reset). Connection: SW1 (STOP) → 8051 MCU INT0 (P3.2) SW4 (RESET) → 8051 MCU INT1 (P3.3)
4	7-segment display	Shows the number of completed cycles (0-10). Connection: Data to P2, Control to P3.6 & P3.7
5	LCD display	Displays menus, modes, and system states. Connection: Data to P0, Control to P3.0, P3.1, P3.4
6	Servo motor	Rotates back and forth based on the selected speed's PWM. Connection: PWM signal to P3.5

Figure 2.1.2 Table of Servo Motor Control Module Hardware Partition

2.2 Software Partition

No.	Module	Description
1	Flag-Based Super Loop	The main continuous loop that manages the LCD redraws, keypad polling, and servo motor execution based on memory flags.
2	Hardware Interrupts	ORG 0003H handles the Emergency Stop by clearing the MOTOR_EN flag. ORG 0013H handles Reset by changing UI_STATE without halting the motor.
3	Non-Blocking Keypad Scan	Rapidly scans the matrix without SJMP delays, returning #0FFH if no key is pressed so the motor is not blocked.
4	Breakable Motor Delays	Executes PWM pulses using Timer 0. Checks the MOTOR_EN and UI_DIRTY flags during the delay loop to instantly abort the delay if an interrupt was triggered.
5	7-Segment Multiplexing	Divides CYCLE_COUNT by 10 to isolate digits, updating DIS0 and DIS1 rapidly.
6	LCD Manager	Updates the screen dynamically based on the UI_STATE flag. Prevents flickering by only drawing when UI_DIRTY is set to 1.

Table 2.2 Table of Servo Motor Control Module Software partition

3.0 Algorithm

3.1 Declaration

Name	Memory Address / Port	Function
LCD_DAT	P0	8-bit data port for the LCD
LCD_EN	P3.0	LCD Enable pin
LCD_RW	P3.1	LCD Read/Write pin
LCD_RS	P3.4	LCD Register Select pin
KEY_PORT	P1	Port connected to the 4x4 hexadecimal keypad
SERVO_PIN	P3.5	PWM output pin to control the servo motor
SEG_PORT	P2	8-bit data port for 7-segment display
DIS0	P3.6	Selects digit 0 of the 7-segment display
DIS1	P3.7	Selects digit 1 of the 7-segment display
CYCLE_COUNT	31H	Tracks the number of completed motor cycles (0 to 10)
SPEED_MODE	32H	Stores the confirmed, active speed mode (0-3)
TEMP_SPEED	33H	Temporarily stores the selected speed until confirmed
UI_STATE	35H	Determines the current screen (0=Menu, 1=Confirm, 2=Running, 3=Stop, 4=Complete)
MOTOR_EN	36H	Master flag (1 = Motor running, 0 = Motor off)
UI_DIRTY	37H	Flag to trigger an LCD redraw (1 = Redraw needed, 0 = LCD is current)

Table 3.1 Description of Algorithm name

3.2 Pseudocode

3.2.1 Main Module

START

Initialize INT0, INT1 (falling edge), and Timer 0 (Mode 1)

Initialize variables: CYCLE_COUNT = 0, UI_STATE = 0, MOTOR_EN = 0

Call INIT_LCD and STARTUP_SEQ

SUPER_LOOP:

// 1. UI Screen Update

If UI_DIRTY is 1:

Clear UI_DIRTY

Clear LCD and draw current state (Menu, Confirm, Running, etc.)

// 2. State & Input Handling

If UI_STATE is Menu or Confirm:

Call READ_KEYPAD_FAST to check for user input

Update SPEED_MODE and UI_STATE based on input

// 3. Motor Execution

If MOTOR_EN is 1:

Call SERVO_180 and SERVO_0 using delays based on SPEED_MODE

If movement completes without interrupt, increment CYCLE_COUNT

If CYCLE_COUNT is 10, set MOTOR_EN to 0 and UI_STATE to 4 (Complete)

Call UPDATE_DISPLAY

Repeat SUPER_LOOP

3.2.2 LCD Display Module

INIT_LCD:

Send setup commands (38H, 0CH, 01H, 06H) with required delays

WRITE_STRING:

Loop through data pointer, read characters until '0' is found

Call LCD_DATA for each character

LCD_CMD / LCD_DATA:

Set RS (0 for Command, 1 for Data)

Clear RW to write

Pulse EN (Set to 1, then 0)

Call 2ms delay

3.2.3 Keypad Scanning Module

READ_KEYPAD_FAST:

Send 0F0H to Keypad Port

Read Port; if 0F0H, return 0FFH (No key pressed)

Call 20ms debounce delay

Sequentially send low signals to rows and read columns

Identify pressed row/column intersection

Map intersection to KEY_MAP and return ASCII character

3.2.4 Servo Motor Control Module

SERVO_MOVE (180_DEGREE and 0_DEGREE):

Set PWM Pin to 1 (High)

Load Timer 0 for high duration and wait for TF0 flag

Set PWM Pin to 0 (Low)

Load Timer 0 for low duration and wait for TF0 flag

// Breakable Delay Check

Check MOTOR_EN and UI_DIRTY flags during delays

If flags changed by interrupt, abort movement immediately

3.2.5 Interrupt Module

ISR_INT0 (SW1 - Emergency Stop):

- Set MOTOR_EN to 0 (Stops motor loop)
- Set UI_STATE to 3 (Emergency Stop Screen)
- Set UI_DIRTY to 1 (Trigger screen redraw)
- Return from interrupt

ISR_INT1 (SW4 - Reset System):

- Set UI_STATE to 0 (Main Menu Screen)
- Set UI_DIRTY to 1 (Trigger screen redraw)
- Return from interrupt

3.2.6 7-Segment Display Module

UPDATE_DISPLAY:

- Divide CYCLE_COUNT by 10
- Store remainder as Units and quotient as Tens

- Fetch 7-segment code for Units
- Send to Port, enable Display 1, call 40us delay, disable Display 1

- Fetch 7-segment code for Tens
- Send to Port, enable Display 0, call 40us delay, disable Display 0

3.2.7 Delay Module

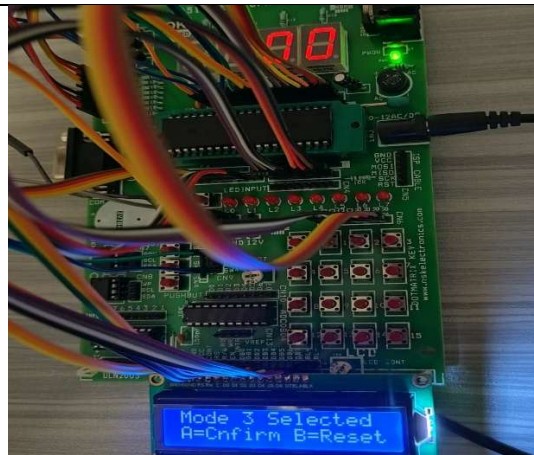
DELAYS (40us, 2ms, 5ms, 20ms, 50ms, 1s):

- Load predefined loop counts into registers (R5, R6, R7)
- Decrement registers in nested loops until zero is reached
- Return

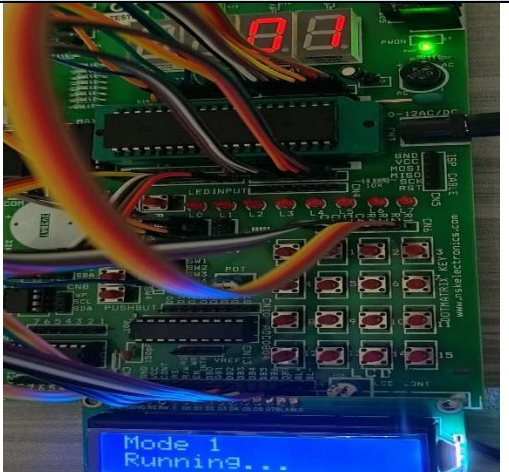
4.0 Snap Pictures of final product



System startup sequence displaying group member names.



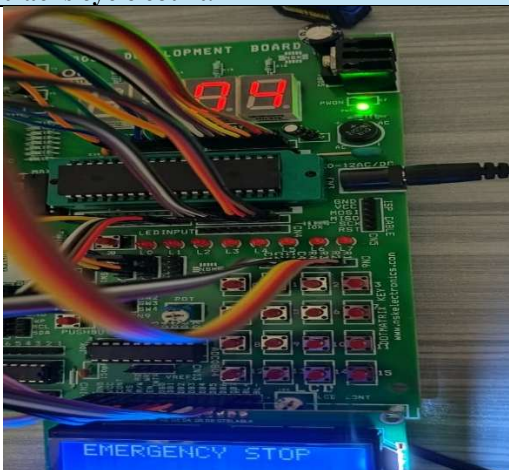
Confirmation prompt for the selected speed mode



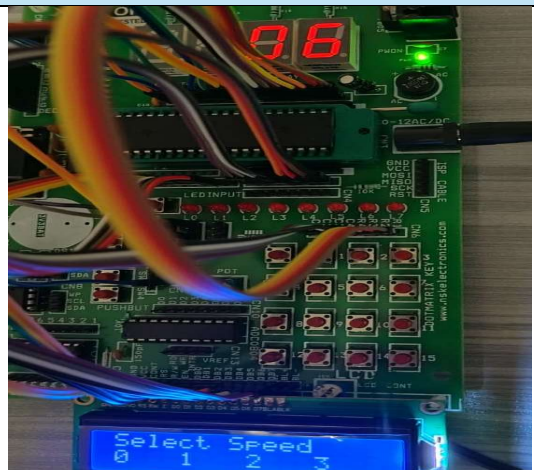
System actively running Mode 1 while 7-segment tracks cycle count.



Correction: System completing 10 motor cycles and displaying the main menu



System halted via hardware interrupt displaying EMERGENCY STOP.



System UI reset to main menu while the motor continues running.

5.0 References

Apoorve. (2025). What is a Servo Motor? - Understanding the basics of Servo Motor Working. Circuit Digest. Available at: <https://circuitdigest.com/article/servo-motor-working-and-basics>.